

The Spectrum of an Operator

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Definition

Let T be a linear operator, $\lambda \in \mathbb{C}$ is called an eigenvalue of T , if there exists $x \in H$ with $x \neq 0$ such that

$$Tx = \lambda x$$

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Example

Consider

$$T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$x \longmapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} x$$

$1 \in \mathbb{C}$ is an eigenvalue of T , since there is $(1, 0) \in \mathbb{R}^2$ such that

$$T \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

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Theorem

Let $\mathcal{H}$ be a Hilbert space and $T : \mathcal{H} \rightarrow \mathcal{H}$ be bounded linear then T is invertible $\iff \ker T^ = \{0\}$ and $\exists \alpha > 0$ such that $\|T(x)\| \geq \alpha \|x\| \quad \forall x \in \mathcal{H}$.*

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Corollary

Let $\mathcal{H}$ be a Hilbert space and $T : \mathcal{H} \rightarrow \mathcal{H}$ be bounded linear then T is invertible $\iff \exists \beta > 0$ such that $\|T^(x)\| \geq \beta \|x\|$ and $\exists \alpha > 0$ such that $\|T(x)\| \geq \alpha \|x\| \quad \forall x \in \mathcal{H}$.*

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Proof.

If T is invertible, then so is T^* . So, by Theorem 1, $\exists \beta > 0$ such that $\|T^*(x)\| \geq \beta \|x\|$ and $\exists \alpha > 0$ such that $\|T(x)\| \geq \alpha \|x\|$ for all $x \in H$. Conversely, if $\exists \beta > 0$ such that $\|T^*(x)\| \geq \beta \|x\|$, then if $T^*u = 0$ for some $u \in H$, then

$$\begin{aligned} 0 &= \|T^*u\| \geq \beta \|u\| \\ &\implies u = 0 \end{aligned}$$

So, $\ker T^* = \{0\}$. By Theorem 1, T is invertible. □

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Definition

A linear operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is said to be **normal** if $TT^* = T^*T$.

Definition

A linear operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is said to be **self-adjoint** if $T = T^*$.

Definition

A linear operator $U : \mathcal{H} \rightarrow \mathcal{H}$ is said to be **unitary** if $UU^* = U^*U = I$

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Definition

Let T be a linear operator. The resolvent set of T , denoted by $\rho(T)$, is $\rho(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is invertible}\}$. The spectrum of T , denoted by $\sigma(T)$ is $\sigma(T) = \mathbb{C} \setminus \rho(T)$, or equivalently, $\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible}\}$

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Remark

Notice that, if $T \in B(\mathcal{H})$, by Theorem 1, we can rewrite the resolvent set of T as

$$\rho(T) = \{\lambda \in \mathbb{C} : \ker(T - \lambda I)^* = \{0\}\}$$

$$\text{and } \|(T - \lambda I)x\| \geq \alpha \|x\| \text{ for some } \alpha > 0\}$$

Alternatively, we can rewrite the resolvent set of T as

$$\rho(T) = \{\lambda \in \mathbb{C} : \|(T - \lambda I)^*x\| \geq \beta \|x\|\}$$

$$\text{and } \|(T - \lambda I)x\| \geq \alpha \|x\| \text{ for some } \beta, \alpha > 0\}$$

by Corollary 1

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Theorem

If λ is an eigenvalue of T , then $\lambda \in \sigma(T)$

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Proof.

Suppose that $\lambda \notin \sigma(T)$, then $T - \lambda I$ is invertible. So, we have $(T - \lambda I)u = 0 \implies u = (T - \lambda I)^{-1}(T - \lambda I)u = (T - \lambda I)^{-1}(0) = 0$. So, if there is $u \in \mathcal{H}$ such that $Tu = \lambda u$, then $u = 0$. So, λ is not an eigenvalue of T . $\square$

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Theorem

Let $\mathcal{H}$ be a finite dimensional Hilbert space, and let $T : \mathcal{H} \rightarrow \mathcal{H}$ be linear. If $\lambda \in \sigma(T)$, then λ is an eigenvalue of T .

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Remark

For finite dimensional Hilbert space, the spectrum of T consist only the eigenvalues of T . However, in infinite dimensional Hilbert space, we may have no eigenvalues. For instance, consider the shift operator. Here, it has no eigenvalues, since if we consider $(T - \lambda I)x = 0$, we see that

$$\begin{aligned}(-\lambda x_1, x_1 - \lambda x_2, x_2 - \lambda x_3, \dots) &= (0, 0, 0, \dots) \\ \implies x &= (0, 0, 0, \dots)\end{aligned}$$

So, T does not have an eigenvalue. However, we showed that T is not invertible. So, $T - 0I$ is not invertible and $0 \in \sigma(T)$. So, the spectrum of an operator does not only consist of eigenvalues.

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Proof.

Suppose that λ is not an eigenvalue of T , then there is nonzero $u \in \mathcal{H}$ such that $Tu = \lambda u$. So, there is no nonzero $u \in \mathcal{H}$ such that $(T - \lambda I)u = 0$. So, $\ker(T - \lambda I) = \{0\}$. By rank-nullity theorem, we have

$$\begin{aligned}\dim \mathcal{H} &= \dim \ker(T - \lambda I) + \dim \operatorname{Im}(T - \lambda I) \\ &= 0 + \dim \operatorname{Im}(T - \lambda I) \\ &= \dim \operatorname{Im}(T - \lambda I)\end{aligned}$$

So, $\operatorname{Im}(T - \lambda I) = \mathcal{H}$. This means that $(T - \lambda I)$ is surjective, and therefore, invertible. So, $\lambda \in \rho(T)$, and therefore, $\lambda \notin \sigma(T)$. □

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Proposition

Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be linear operator. The following properties hold:

$$\textcircled{1} \quad \sigma(I) = \{1\}$$

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Proposition

Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be linear operator. The following properties hold:

- ① $\sigma(I) = \{1\}$
- ② $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$

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Proposition

Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be linear operator. The following properties hold:

- ❶ $\sigma(I) = \{1\}$
- ❷ $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
- ❸ *If T is invertible, then $\sigma(T^{-1}) = \{\mu^{-1} : \mu \in \sigma(T)\}$*

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Proposition

Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be linear operator. The following properties hold:

- ❶ $\sigma(I) = \{1\}$
- ❷ $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
- ❸ *If T is invertible, then $\sigma(T^{-1}) = \{\mu^{-1} : \mu \in \sigma(T)\}$*
- ❹ *If T is self-adjoint, then $\sigma(T) \subseteq \mathbb{R}$.*

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Proposition

Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be linear operator. The following properties hold:

- ① $\sigma(I) = \{1\}$
- ② $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
- ③ *If T is invertible, then $\sigma(T^{-1}) = \{\mu^{-1} : \mu \in \sigma(T)\}$*
- ④ *If T is self-adjoint, then $\sigma(T) \subseteq \mathbb{R}$.*
- ⑤ *If $U : \mathcal{H} \rightarrow \mathcal{H}$ is invertible, then $\sigma(UTU^{-1}) = \sigma(T)$*

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Proof.

- **Proof of 1:** Suppose that $\lambda \in \sigma(I)$, then $I - \lambda I = (1 - \lambda)I$ is not invertible. $(1 - \lambda)I$ is not invertible if and only if $1 - \lambda = 0$, since $1 - \lambda \in \mathbb{F}$. So, this is true if and only if $\lambda = 1$.

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Proof.

- **Proof of 1:** Suppose that $\lambda \in \sigma(I)$, then $I - \lambda I = (1 - \lambda)I$ is not invertible. $(1 - \lambda)I$ is not invertible if and only if $1 - \lambda = 0$, since $1 - \lambda \in \mathbb{F}$. So, this is true if and only if $\lambda = 1$.
- **Proof of 2:** Suppose that $\mu \in \sigma(T^*)$, then $T^* - \mu I$ is not invertible. Observe that $T^* - \mu I$ is invertible if and only if $(T^* - \mu I)^* = T - \bar{\mu}I$ is invertible, since if $(T^* - \mu I)^*$ is invertible, then $((T^* - \mu I)^*)^* = T^* - \mu I$ is invertible. So, $T^* - \mu I$ is not invertible iff $(T^* - \mu I)^* = T - \bar{\mu}I$ is not invertible. So, $\bar{\mu} \in \sigma(T)$, i.e. $\mu \in \{\bar{\lambda} : \lambda \in \sigma(T)\}$.



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Proof.

- **Proof of 3:** If T is invertible, i.e. $T - 0I$ is invertible, then $0 \notin \sigma(T)$. Similarly, $0 \notin \sigma(T^{-1})$. Suppose that $\mu \in \sigma(T^{-1})$, then $T^{-1} - \mu I$ is not invertible. Observe that $T^{-1} - \mu I$ is invertible if and only if $T(T^{-1} - \mu I) = I - \mu T$ is invertible, since if $T(T^{-1} - \mu I)$ has an inverse, K , then KT would be an inverse of $T^{-1} - \mu I$. So, $T^{-1} - \mu I$ is not invertible iff $T(T^{-1} - \mu I) = I - \mu T$ is not invertible. Also, $I - \mu T$ is not invertible iff $T - \mu^{-1}I$, where μ^{-1} exists, since $\mu \neq 0$. So, $\mu^{-1} \in \sigma(T)$, i.e. $\mu \in \{\lambda^{-1} : \lambda \in \sigma(T)\}$.

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Proof.

- **Proof of 3:** If T is invertible, i.e. $T - 0I$ is invertible, then $0 \notin \sigma(T)$. Similarly, $0 \notin \sigma(T^{-1})$. Suppose that $\mu \in \sigma(T^{-1})$, then $T^{-1} - \mu I$ is not invertible. Observe that $T^{-1} - \mu I$ is invertible if and only if $T(T^{-1} - \mu I) = I - \mu T$ is invertible, since if $T(T^{-1} - \mu I)$ has an inverse, K , then KT would be an inverse of $T^{-1} - \mu I$. So, $T^{-1} - \mu I$ is not invertible iff $T(T^{-1} - \mu I) = I - \mu T$ is not invertible. Also, $I - \mu T$ is not invertible iff $T - \mu^{-1}I$, where μ^{-1} exists, since $\mu \neq 0$. So, $\mu^{-1} \in \sigma(T)$, i.e. $\mu \in \{\lambda^{-1} : \lambda \in \sigma(T)\}$.
- **Proof of 5:** Suppose that $\lambda \in \sigma(UTU^{-1})$, then $UTU^{-1} - \lambda I$ is not invertible. Observe that $UTU^{-1} - \lambda I = U(T - \lambda)U^{-1}$ is not invertible iff $T - \lambda I$ is not invertible. So, $\lambda \in \sigma(T)$.

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Example

Quantum Physics We have seen that for an observable t , we associate a self-adjoint operator T . **The eigenvalues of the operator T represent the possible values of that observable t .**

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Example

Quantum Physics We have seen that for an observable t , we associate a self-adjoint operator T . **The eigenvalues of the operator T represent the possible values of that observable t .** For example, if we want to find the possible values of the energy, we need to find the eigenvalues of H . In other words, we solve

$$H\psi = E\psi = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi.$$

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Example

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$$H\psi = E\psi = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi.$$

*This is the famous **Schrödinger** equation.*

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Theorem

Let $\mathcal{H}$ be a complex Hilbert space and let $T \in B(\mathcal{H})$. If $|\lambda| > \|T\|$ then $\lambda \notin \sigma(T)$.

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Proof.

Let $u \in \ker(T - \lambda I)^*$, then $(T - \lambda I)^*u = (T^* - \bar{\lambda}I)u = 0$. So,
 $T^*u - \bar{\lambda}u = 0 \implies T^*u = \bar{\lambda}u$. So,

$$|\lambda|\|u\| = |\bar{\lambda}|\|u\| = \|\bar{\lambda}u\| = \|T^*u\| \leq \|T^*\|\|u\| = \|T\|\|u\|$$

$$\implies (|\lambda| - \|T\|)\|u\| \leq 0$$

Since $|\lambda| > \|T\|$, then $\|u\| \leq 0 \implies \|u\| = 0 \implies u = 0$.

Hence, $\ker(T - \lambda I)^* = \{0\}$. □

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Proof.

We also have that

$$\begin{aligned}\|(T - \lambda I)x\| &= \|Tx - \lambda x\| \geq \left| \|Tx\| - |\lambda| \|x\| \right| \\ &= \left| \frac{\|Tx\|}{\|x\|} - |\lambda| \right| \|x\| \\ &= \left(|\lambda| - \frac{\|Tx\|}{\|x\|} \right) \|x\| \\ &\geq (|\lambda| - \|T\|) \|x\| \\ &= \alpha \|x\|\end{aligned}$$

where $\alpha := |\lambda| - \|T\|$ is positive, since $|\lambda| > \|T\|$. So, $\lambda \in \rho(T)$, by first equality in Remark 1. Hence, $\lambda \notin \sigma(T)$. $\square$

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Corollary

If $T \in B(\mathcal{H})$, then $\sigma(T)$ is bounded.

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Proof.

If $\lambda \in \sigma(T)$, then, by the contrapositive of Theorem 4, $|\lambda| \leq \|T\|$. So, $\lambda \in \overline{B(0, \|T\|)}$. Hence, $\sigma(T) \subseteq \overline{B(0, \|T\|)}$, i.e. $\sigma(T)$ is bounded. □

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Corollary

If $\mathcal{H}$ is a complex Hilbert space and $U \in B(\mathcal{H})$ is unitary then

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

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Proposition

If $\mathcal{H}$ is a complex Hilbert space and $U \in B(\mathcal{H})$ is unitary then

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

Proof.

If $|\lambda| > 1 = \|U\|$, then $\lambda \notin \sigma(U)$. Moreover, if $|\lambda| < 1$, then

$$U - \lambda I = -\lambda U(U^{-1} - \frac{1}{\lambda}I)$$

is invertible, since $-\lambda U$ is invertible and $\left|\frac{1}{\lambda}\right| > 1 = \|U^{-1}\|$. So, $\lambda \notin \sigma(U)$. Therefore, $|\lambda| = 1$ if $\lambda \in \sigma(U)$. □

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Theorem

If $T \in B(\mathcal{H})$, then $\sigma(T) \subseteq \mathbb{C}$ is a closed set.

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Proof.

We will show that $\rho(T) = \mathbb{C} \setminus \sigma(T)$ is open. Let $\lambda \in \rho(T)$. This means that, by second equality in Remark 1, there is $\alpha > 0$ such that

$$\|(T - \lambda I)u\| \geq \alpha \|u\| \implies \frac{\|(T - \lambda I)u\|}{\|u\|} \geq \alpha$$

Since the quantity $\frac{\|(T - \lambda I)u\|}{\|u\|}$ has a lower bound, its infimum exists. Similarly, by second equality in Remark 1, the infimum of the quantity $\frac{\|(T - \lambda I)^*u\|}{\|u\|}$ exists. □

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Proof.

Now, let

$$\gamma := \inf_{u \in H \setminus \{0\}} \frac{\|(T - \lambda I)u\|}{\|u\|} \quad \gamma' := \inf_{u \in G \setminus \{0\}} \frac{\|(T - \lambda I)^*u\|}{\|u\|}$$

and let $\delta := \frac{1}{2} \min(\gamma, \gamma')$. We want to show that

$$B(\lambda, \delta) \subseteq \rho(T)$$



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Proof.

Let $\tilde{\lambda} \in B(\lambda, \delta)$, then $|\tilde{\lambda} - \lambda| < \delta$. First, we observe that

$$\begin{aligned}\|(T - \tilde{\lambda}I)u\| &= \|(T - \lambda I)u - (\tilde{\lambda} - \lambda)I(u)\| \\ &\geq \left| \|(T - \lambda I)u\| - |\tilde{\lambda} - \lambda| \|u\| \right| \\ &= \left| \frac{\|(T - \lambda I)u\|}{\|u\|} - |\tilde{\lambda} - \lambda| \right| \|u\| \\ &= \left(\frac{\|(T - \lambda I)u\|}{\|u\|} - |\tilde{\lambda} - \lambda| \right) \|u\| \\ &> \left(\gamma - \frac{1}{2} \min(\gamma, \gamma') \right) \|u\| \\ &\geq \left(\gamma - \frac{\gamma}{2} \right) \|u\| \\ &= \frac{\gamma}{2} \|u\|\end{aligned}$$

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Proof.

Similarly, we have

$$\begin{aligned}\|(T - \tilde{\lambda}I)^*u\| &= \|(T - \lambda I)^*u - \overline{(\tilde{\lambda} - \lambda)}I(u)\| \\ &\geq \| |(T - \lambda I)^*u| - |\overline{\tilde{\lambda} - \lambda}| \|u\| \\ &= \| |(T - \lambda I)^*u| - |\tilde{\lambda} - \lambda| \|u\| \\ &= \left| \frac{\|(T - \lambda I)^*u\|}{\|u\|} - |\tilde{\lambda} - \lambda| \right| \|u\| \\ &= \left(\frac{\|(T - \lambda I)^*u\|}{\|u\|} - |\tilde{\lambda} - \lambda| \right) \|u\| \\ &> (\gamma' - \frac{1}{2} \min(\gamma, \gamma')) \|u\| \geq (\gamma' - \frac{\gamma'}{2}) \|u\| = \frac{\gamma'}{2} \|u\|\end{aligned}$$

So, by the second equality in Remark 1, $\tilde{\lambda} \in \rho(T)$. Therefore,
 $B(\lambda, \delta) \subseteq \rho(T)$.



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Corollary

If $T \in B(\mathcal{H})$, then $\sigma(T)$ is compact in $\mathbb{C}$

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Example

We would like to find the spectrum of the unilateral shift, $T : \ell^2 \rightarrow \ell^2$, defined by

$$T(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$$

Since T has been shown to have no eigenvalue, we will consider the eigenvalues of T^ instead and use the properties of the spectrum to have a lower bound for the spectrum.*

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Suppose that $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$, then consider the sequence

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Example

Suppose that $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$, then consider the sequence

$$x = (1, \lambda, \lambda^2, \dots)$$

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Claim: $x \in \ell^2$

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Suppose that $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$, then consider the sequence

$$x = (1, \lambda, \lambda^2, \dots)$$

Claim: $x \in \ell^2$

We have

$$\sum_{n=0}^{\infty} |x_n|^2 = \sum_{n=0}^{\infty} (|\lambda|^n)^2 = \sum_{n=0}^{\infty} (|\lambda|^2)^n = \frac{1}{1 - |\lambda|^2} < \infty$$

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Now, observe that

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Example

Now, observe that

$$\begin{aligned} T^*x &= T^*(1, \lambda, \lambda^2, \dots) \\ &= (\lambda, \lambda^2, \lambda^3, \dots) = \lambda(1, \lambda, \lambda^2, \dots) = \lambda x \end{aligned}$$

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So, every $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$ is an eigenvalue of T^ .*

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Now, observe that

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So, every $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$ is an eigenvalue of T^ . Now, we have*

$$\{\lambda : |\lambda| < 1\} = \{\bar{\lambda} : |\lambda| < 1\} \subseteq \sigma(T)$$

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So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

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Example

So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

or

$$B(0, 1) \subseteq \sigma(T) \subseteq \overline{B(0, 1)}$$

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Example

So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

or

$$B(0, 1) \subseteq \sigma(T) \subseteq \overline{B(0, 1)}$$

Since $\sigma(T)$ is closed, we have

$$\sigma(T) = \overline{B(0, 1)} = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$